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Improved coot optimizer algorithm-based MPPT for PV systems under complex partial shading conditions and load variation

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ABSTRACT

Solar power is considered one of the most common renewable energy sources. However, the effective harnessing of maximum solar energy in photovoltaic (PV) systems faces a significant challenge due to weather fluctuations. This challenge becomes particularly pronounced for PV systems aiming to achieve optimal power output during Maximum Power Point Tracking (MPPT), especially under partial shading conditions (PSCs). In the P-V curve of PV arrays, the PSCs cause multiple peaks, known as local peaks, and one global peak. The use of sophisticated MPPT optimization algorithms is necessary to identify the global peak, ensuring the highest point of power production and avoiding entrapment in the local peaks. The main drawbacks of these optimization algorithms are their inability to differentiate between the irradiance change and load variations, as well as their high convergence speed. This paper proposes an Improved Coot Optimization Algorithm (ICOA) based on MPPT to alleviate the issue of convergence speed. Furthermore, a novel approach has also been devised to enhance the speed response during load variation, which can be implemented with any DC-DC converter. A novel approach was adopted, with one tuning parameter implemented to simplify the algorithm. A variety of complex PSCs were tested with a SEPIC converter, and the sampling time was adjusted at 0.05 s. Based on the experimental results, the proposed ICOA has achieved the best performance, with an average tracking time of 0.58 s under different weather conditions and an efficiency of 99.94 %. Additionally, an assessment of the proposed technique against existing metaheuristic algorithms in this field is conducted, revealing the effectiveness of the proposed approach in terms of rapid tracking and high efficiency.

Introduction

The growing concerns about global warming, the diminishing availability of fundamental energy sources like coal, and rising fossil fuel prices have naturally drawn the expertise of professionals toward Renewable Energy Sources (RES) [1]. Throughout the day, PV systems are exposed to diverse weather conditions that might influence their efficiency. To address this, the PV systems are equipped with the Maximum Power Point Tracking (MPPT) method to increase overall PV efficiency [2–4]. MPPT represents a crucial methodology within PV systems. It consistently adjusts the operational settings of solar panels to guarantee they operate at their optimal efficiency, ultimately maximizing energy generation. This subject is a prominent area of research, with a multitude of MPPT algorithms proposed in the existing body of literature [5]. Several techniques have been employed to improve the effectiveness of PV modules, such as hill climbing (HC) [6], incremental conductance (IC) [7], and perturb and observe (P&O) [8]. However, conventional methods demonstrated the ability to track the maximum power under USCs with a high convergence speed and oscillation around the steady-state point. Moreover, they are not effective when multiple peaks occur at the output under partial shading conditions (PSCs). The occurrence of multiple peaks in the output power-voltage (P-V) curve,

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observed under such conditions, presents a notable challenge.

To tackle the MPPT issue in PSCs, scientists have implemented advanced soft computing techniques, including fuzzy logic control (FLC) [9] and artificial neural networks (ANN) [10]. These enhanced intelligent strategies efficiently address the constraints associated with conventional methods. However, the success of these methods is hindered by the challenging process of training on large datasets, the requirement of prior knowledge about the PV system, the handling of controlled variables, and the use of fixed membership functions. In addition, these factors lead to substantial memory demands, adding complexity to their implementation. Modern MPPT algorithms, which include both swarm-based and evolutionary algorithms and are frequently bio-inspired, are favored due to their higher steady-state efficiency [11]. Once they have located the GMPP, metaheuristic techniques excel at stabilizing the system, and reducing oscillations while maintaining optimal operation. Furthermore, they enhance efficiency by narrowing their exploration within the P-V or I-V search space, concentrating on the most favorable regions for tracking the GMPP [11–14]. This efficient approach minimizes computational demands and enhances the stability and speed of MPPT algorithms. Moreover, they prevent the necessity for preexisting knowledge of PV system parameters and the frequently complex adjustment of Proportional-Integral-Derivative (PI/PID) controllers, as they directly regulate the duty cycle. In addition, the selection of converter type has minimal effect on implementing these metaheuristic algorithms. However, certain challenges and areas for improvement exist. One notable challenge is scalability. Metaheuristic algorithms typically involve several tuning parameters that need to be set offline, and this process can be time-consuming and complex. The number of parameters necessitating tuning may vary depending on the magnitude of the photovoltaic array. For instance, particle swarm optimization (PSO) [13] requires tuning of three parameters (*C1, C2,* and *w*). In comparison, the firework algorithm [14] requires tuning of five parameters (*A, a, b, m, £*), the cuckoo search algorithm (CS) [15] requires tuning of three parameters (P_a, β, α), the memetic salp swarm algorithm (MSSA) [16] requires tuning of three parameters (*n, M,* k_{max}) and the improved squirrel search algorithm (ISSA) [17] requires tuning of two parameters (P_r, d_g).

Furthermore, random number generators introduce randomness to facilitate effective search space exploration. However, excessive reliance on random numbers, especially when updating parameters for tasks like tracking the MPP in PV systems, can lead to issues. For instance, in the grey wolf optimizer (GWO) algorithm [18], the C parameter is adjusted by multiplying it with a random integer. It may cause the algorithm to converge toward the local maximum power point (LMPP) and hinder further search space exploration. Striking a balance between randomness and deterministic factors in parameter updates is essential to prevent overreliance on random numbers and improve the algorithm's efficiency in finding the GMPP.

Numerous metaheuristic algorithms, like moth flame optimization (MFO) [19], bat algorithm (BA) [20], whale optimization (WO) [21], and energy valley optimizer (EVO) [22] encounter a common limitation. Specifically, these algorithms cannot distinguish between the irradiance change and load variations. Consequently, when the system encounters a disturbance, the algorithm will initiate a re-initialization process to follow the new maximum point, regardless of whether the variations in power stem from changes in irradiance or load fluctuations, which leads to an increase in the tracking time.

Moreover, a prominent challenge within bio-inspired MPPT algorithms pertains to the need for re-initialization of particles in response to variations in input power (P_{PV}) after the detection of the Global Maximum Power Point (GMPP) represented by P_{MPP}. This challenge is notably observed in several algorithms, including but not limited to ant colony [23], bat [24], flower pollination algorithm (FPA) [25], grasshopper optimization (GHO) [26], salp swarm optimization (SSA) [27], and previously mentioned metaheuristic approaches. Various techniques were developed to differentiate between the irradiance change and load variation and to prevent re-initializing algorithms during load variations. In [28,29], an adaptive tracking method was proposed for systems experiencing load variations, employing an adaptive step-size-based perturbation (ΔD) to track back the maximum point without the need to re-initialize the algorithm. The drawback of this method is that if the step size is increased, it will provide fast tracking but may generate oscillations, whilst diminishing the step size allows for accurate tracking with reduced oscillation but at the expense of prolonging the tracking time. The Improved Differential Evolution (IDE) technique [30] eliminates the necessity of re-initializing particles when there are load variations. Instead, it employs the impedance conversion equation that is specific to the DC-DC converter to retrace its steps back to GMPP [31]. Nevertheless, if the converter has been changed, the equations must change as well. However, these suggested methods have drawbacks like lengthy training times, a need for computational resources, and the requirement for parameter tuning. Table 1 presents a comprehensive comparison of the benchmark optimization approaches and the newly suggested ICOA. The comparison takes into account aspects such as convergence speed, tracking efficiency, oscillations in a steady state, and the number of tuning parameters.

Table 1
A comprehensive comparison between Swarm Intelligence Algorithms MPPT Techniques.

MPPT algorithm	convergence speed	Steady-state oscillation	Tracking efficiency	No. of control parameters
ICOA	Very High	Very Low	Very High	1
GWO [18]	High	Medium	High	3
FWA [14]	High	Medium	High	3
PSO [13]	Medium	High	Medium	3
MSSO [16]	Medium	High	High	3
MFO [19]	Medium	Low	High	3
WO [21]	Medium	Low	Medium	5
GHO [26]	High	Low	High	3
EVO [22]	Very High	Low	High	1
IDE [30]	High	Medium	Medium	3
ISO [32]	High	High	High	2
ISSA [17]	High	Low	High	2
FPA [25]	Medium	Medium	High	3
BA [20]	High	Medium	High	3
ITG [28]	High	Medium	Medium	1
SSA [27]	High	Low	Medium	3
CS [15]	Medium	Low	High	3

The main contributions outlined in this paper are enumerated as follows:

1. Introducing a novel algorithm called ICOA as a new MPPT controller for PV systems to achieve fast and accurate tracking with heightened efficiency. Additionally, only one tuning parameter, accompanied by a novel strategy to control parameters, is employed.
2. Implementing an experimental setup for a PV system incorporating the MPPT mechanism using five benchmark optimization algorithms for comparative evaluation with the newly proposed ICOA
3. A novel adaptive perturbation step-size strategy that minimizes oscillations and is adaptable to a variety of converters has been proposed for rapidly changing load conditions.

The succeeding segments of this paper are organized as follows: Section 2 comprehensively discusses the modeling of partially shaded PV systems. Section 3 provides an overview of the standard Coot Optimization approach; Section 4 details the development of ICOA. Sections 5 and 6 delve into the distinction between varying irradiance and the proposed approach for load variations, respectively. In Section 7, experimental results are discussed, and in Section 8, the proposed method is subjected to testing against alternative methods to validate its effectiveness. Finally, Section 9 concludes the entirety of the paper.

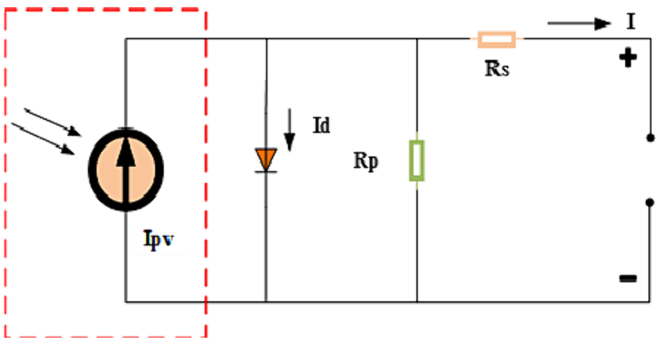


Fig. 1. PV cell modeling.

Modeling of PV systems with PSCs

PV model fundamental characteristics

The PV structure, shown in Fig. 1, involves critical components such as a current source, diode, series resistor (R_s), and a parallel shunt resistor (R_{sh}). The single-diode representation describes the mathematical model used for the PV cell. In the model mentioned above, the PV cell current is defined using Eq. (1).

$$I_{pv} = I_{ph} - I_o \times \left(e\left(\frac{q(V_{pv} + R_s{*}I_{pv})}{akT} - 1\right)\right) - \frac{(V_{pv} + R_s{*}I_{pv})}{R_{sh}} \tag{1}$$

where I_{ph} is the current source, I_{pv} and V_{pv} are the voltage and current of the PV, k is the Boltzmann constant, T is the temperature in degrees, and R_s and R_{sh} are the ohmic values of the series and parallel resistors.

Numerous material characteristics affect the photocurrent I_{ph} of a solar cell. But it may be roughly estimated with enough accuracy like linear dependent upon both irradiance and temperature as shown in Eq. (2):

$$I_{ph} = \frac{G}{G_{ref}}\left[I_{sc,ref} + \mu_{sc}\left(T - T_{ref}\right)\right] \tag{2}$$

Where $I_{sc,ref}$ denotes short-circuit current for the solar cell: G_{ref} is 1000 W/m^2, and T_{ref} is 25 °C. Actual irradiance intensity is G (measured in W/m^2).

This equation is used to estimate diode saturation currents following in Eq. (3):

$$I_o = I_{o,ref}\left(\frac{T_{ref}}{T}\right)^3 e\left[\frac{q}{nk}\left(\frac{1}{T_{ref}} - \frac{1}{T}\right)\right] \tag{3}$$

Where Eq. (4) determines the current nominally saturated $I_{o,ref}$.

$$I_{o,ref} = \frac{I_{sc,ref}}{e\left(\frac{V_{oc,ref}}{nV_t}\right) - 1} \tag{4}$$

Where the $V_{oc,ref}$ is the Open circuit voltage for the reference.

The effects of PSCs on MPPT

The imperative for PV systems to consistently operate at their maximum output point stems from their inherent characteristic of possessing low efficiency. Within a conventional PV system, solar cells are organized into modules and arrays interconnected either in series or parallel configurations. The occurrence of PSC can potentially diminish the output and voltage of specific PV modules. To mitigate this, parallel bypass diodes are strategically incorporated for continuous electricity flow within each module. However, bypass diodes are essential to prevent hotspots on the surface of solar cells resulting from module mismatches [33]. Furthermore, blocking diodes are employed at series ends to prevent reverse current flow. MPPT algorithms are designed to adapt

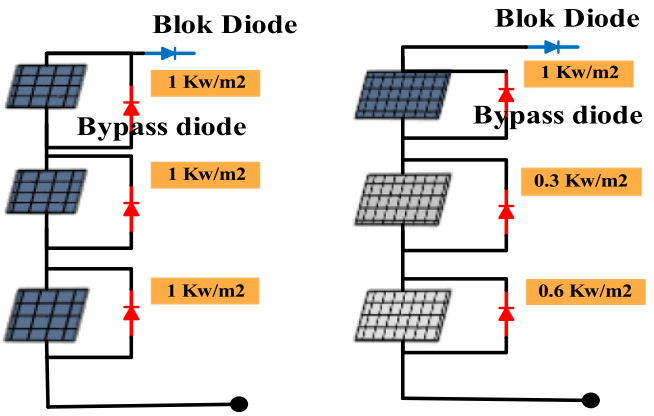


Fig. 2. Irradiance scenarios for uniformity at STC (a) and PS condition (b).

to PSC, facilitating the identification of the GMPP for comprehensive system optimization. The visual representation in Fig. 2 illustrates a configuration featuring a bypass diode within a series-connected set of three PV modules. Each PV module receives the same amount of irradiance under similar conditions, and the voltage is unaffected. The output of a PV system under uniform irradiation is shown in Fig. 3 (a) by the (P-V) and (I-V) curves. When the PV modules are partially shaded, as in Fig. 3 (b), the P-V and I-V curves under non-uniform irradiance have many local maxima (LMs) and one global maximum (GM). Local maximum power points (LMPPs) are the terms used to describe the other peaks, whereas the highest peak is known as the GMPP. The simulation was implemented within the MATLAB/Simulink 2022a environment, utilizing a personal computing apparatus endowed with an Intel® Core™ i7 CPU operating at 3.4 GHz and 16 GB of RAM. The computational platform operated on the Windows 10 operating system. The solver employed for the simulation was ode45, specifically utilizing the Dormand-Prince method, with an auto variable-step size.

Coot optimization algorithm

In 2021, Iraj Naruei [34] introduced the Coot Optimization Algorithm (COA), a prominent *meta*-heuristic optimization technique inspired by the collective behaviors observed in coot swarms and waterfowl of the Rallidae family. This algorithm simulates a virtual population of coots, iteratively adjusting their positions based on predefined rules that emulate the foraging patterns of coot swarms. The COA algorithm encompasses two distinct modes of avian movement over water surfaces, characterized by erratic and consistent motions during the initial and subsequent phases, respectively.

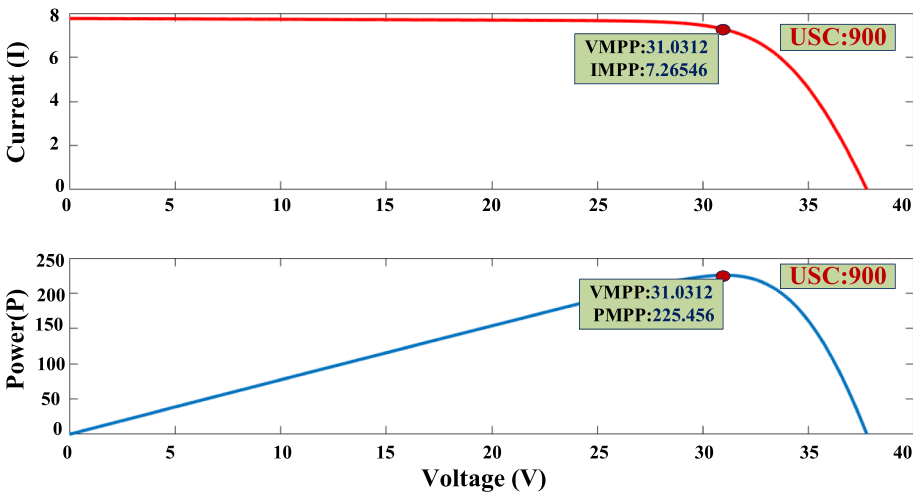
Therefore, four various movements of coots on the water's surface are analyzed, which include:

1. Random motion.
2. Chain motion.
3. Changing the position of the group leaders.
4. Movement of leaders to direct a group to the best location.

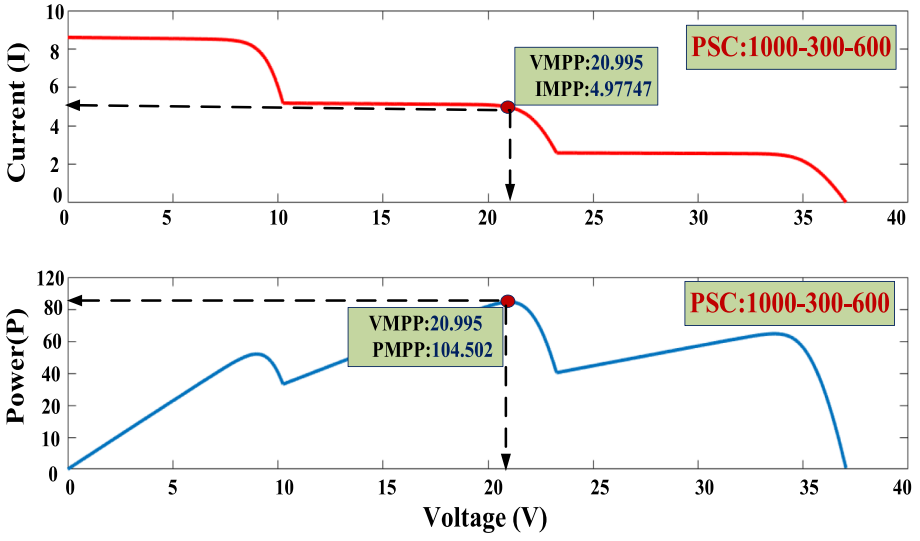
The population in space generated at random can be calculated using Eq. (5):

$$X(k) = rand(1, d) \times (ub - lb) + lb \tag{5}$$

X(k) denotes the location at time *k*, with *d* representing the dimension of the search space (the number of optimization parameters). The upper and lower bounds of the search space are denoted by *ub* and *lb*, respectively. These definitions characterize the four distinct motion phases:



(a)



(b)

Fig. 3. Graphs depicting the P-V and I-V curves under two conditions: (a) Uniform shading and (b) Partial shading.

Random motion

The coot particle should be moved randomly in the search space using Eq. (6) to represent this movement.

$$W = rand(1, d) \times (ub - lb) + lb \tag{6}$$

The coots' movement allows them to investigate various areas in the search space, and this process will allow the algorithm to escape from all local optimums that it may become stuck in. Since the algorithm moved away from these local points, Eq. (7) is used to find the new coot positions.

$$X(k) = X(k) + Z \times R_2 \times (W - X(k)) \tag{7}$$

Where $R2$ denotes a random number within the range of 0 to 1, and Z is calculated using Eq. (8)

$$Z = 1 - L \times \left(\frac{1}{I_{ter}}\right) \tag{8}$$

The maximum number for iteration is represented by the variable "I_{ter}," while the variable "L" indicates the current iteration.

Chain motion

The chain motion involves first calculating the distance between the two coots, after which one of the coots is moved towards the next coot by approximately half of the vector, as shown in Eq. (9):

$$X(k) = 0.5 \times (X(k-1) + X(k)) \tag{9}$$

Where $X(k-1)$ is the second coot.

Changing the position following the group leaders

Coot modifies its location in this phase by the group leaders. The coot approaches the leader, and there are many methods for choosing the leader in different situations. Since there are several leaders, the coot's location is updated using the average location of the leaders. At the same time, Eq. (10) is used to choose the leader.

$$M = 1 + (kMODNL) \tag{10}$$

Where M, represents the index number of the leader that the coot will choose based on the given mechanism. The current coot is represented by the variable *'k'*, while the number of leaders is indicated by $NL = 1$. *MOD*, Modulus operation, ensuring M remains within the valid range of the leader indicated. Therefore, $X(k)$ must reposition itself to align with the leader identified as '*M.* ' X(k) is tasked with updating its position by leader '*M.*' Eq. (11) is employed to determine the position of the next coot is calculated based on the leader number M.

$$X(k) = G(M) + 2 \times R_1 \times cos(2R\pi) \times (G(M) - X(k)) \tag{11}$$

$X(k)$ is the coot's current location, $G(M)$ is the leader position that has been chosen, R_1 is a random value between 0 and 1, π is the value of 3.14, and R is a random value between (−1,1).

Movement of leaders to direct a group to the best location

During this phase, the coots move closer to where the leaders are located while the leaders shift their positions in search of the best location for the coots to follow them. While the best location for the leaders is determined using Eq. (12).

$$G(k) = \begin{cases} B \times R_3 \times cos(2R\pi) \times (g_{Best} - G(k)) + g_{Best} R4 \prec 0.5 \\ B \times R_3 \times \cos(2R\pi) \times (g_{Best} - G(k)) - g_{Best} R4 \succ 0.5 \end{cases} \tag{12}$$

Eq. (13) defines the variables. R_3 and R_4 as random values within [0, 1], R within the range of [-1, 1], and B, as expressed below:

$$B = 2 - L \times \left(\frac{1}{I_{ter}} \right) \tag{13}$$

Where I_{ter} denotes the maximum iteration, and L represents the current iteration.

Proposed improved Coot-Based MPPT method

This section proposes an improved COA to enhance tracking time and efficiency. The proposed modifications aim to simplify the convergence process to optimize the convergence speed (CS), improve the precision of global maximum identification, differentiate between variations in load and changes in irradiance, and respond to load changes quickly. The procedure of the Improved COA (ICOA-MPPT) is stated as follows:

Modification in initialization for COA

During the initialization of the standard COA, agents are assigned randomly. However, in ICOA-MPPT, four agents are specifically chosen with fixed initial positions of 0.2, 0.4, 0.6, and 0.8 duty cycles to cover the entire PV curve. The MPPT algorithm relies on both CS and tracking efficiency as crucial factors. Expanding the agent count can guarantee enhanced precision in tracking the GMPP, but this will increase the CS. As a result, when determining the optimal number of agents for a given search area, there is a compromise between CS and tracking efficiency. Given that the duty cycle varies from 0.1 to 0.9, having four initial agents evenly distributed over the search space is considered adequate. Implementing a uniform distribution over the full P-V search space enables the efficient determination of the GMPP area for all complicated PSCs. This approach avoids the need to position all four agents above or below a 0.5 duty cycle during the initialization step.

Modification in the randomization phase

The randomization phase is modified by adjusting the random factor R, which is changed from a range of [-1, 1] to [0, 1]. This modification addresses the possibility of the D exceeding its threshold as a result of stochastic negative multiplication, hence increasing the CS.

Modification in the tuning parameter

Tuning parameters in optimization algorithms plays a crucial role in determining convergence time. Identifying suitable parameter values and devising empirical rules for their selection serves to alleviate users from the arduous trial-and-error methods conventionally used for parameter determination, often deemed problem-dependent. To address this challenge, a novel strategy to control parameters is employed. This parameter is crucial since it effectively controls exploration and exploitation trade-offs. It is dynamically reduced throughout the iterations, allowing the COA algorithm to explore the search space and subsequently exploit it initially. Eq. (14) is defined as:

$$A = C \times e^{-\left(\frac{2 \times L}{I_{ter}}\right)^2} \tag{14}$$

Where C is a constant parameter with a default value of 1.5, L represents the current iteration.

and I_{ter} presents the maximum iteration.

Update the coot positions

A modified coot optimization method for MPPT is used to adjust the coot locations when the requirement specified in Eq. (19) is not satisfied. The coot's position is altered from Eq. (11) to Eq. (15) by the probability demand ($P = 0.5$). It is so because the duty cycle, D range is between 0.1 and 0.9, and the first cycles of the multiplication of $G(M)$ with 2 will modify the coots' position outside of the D limit.

$$D_i(k+1) = A \times R_1 \times cos(2 \times \pi \times R) \times (D_{Gbest}(M) - D_i(k)) + D_{Gbest}(M) \tag{15}$$

where $D_i(k)$ denotes the location of the coot,D_{Gbest} indicates the best global position, the variable M, corresponds to the index number of the leader that the coot will choose, depending on the specified process outlined in Eq. (10).

The random search space is improved by implementing Eq. (16) and Eq. (17). While the best position $D_{Gbest(k)}$ of the particles replaces the random position (W).

$$W = D_{Gbest}(k \tag{16}$$

$$D_i(k+1) = D_i(k) + A \times abs(D_{Gbest}(k) - D_i(k)) \tag{17}$$

where $D_i(k)$, indicates the location of a coot,$D_{Gbest}(k)$ represents the best global position, while the Chain motion in Eq. (9) is modified to Eq. (18);

$$D_i(k+1) = 0.5 \times (D_i(k-1) + D_i(k)) \tag{18}$$

Modification of the leaders' movement

The primary objective of the leading movement is to guide the coots towards an optimal area. This involves the coots adjusting their positions to align with the leaders' main position, while the leaders themselves also reposition to find an ideal spot that naturally influences the coots' movements. Nevertheless, it is crucial to acknowledge that employing the leading movement approach can introduce delays in the tracking process as the coots adapt to follow the leaders. As a result, this may prolong the overall tracking time compared to MPPT techniques that do not involve such coordination. Removing the leading movement had the potential to enhance tracking speed, enabling the MPPT system to swiftly respond to environmental changes and reach the MPP more efficiently.

Modification in the termination condition

Two termination conditions are defined in condition (19) to put an

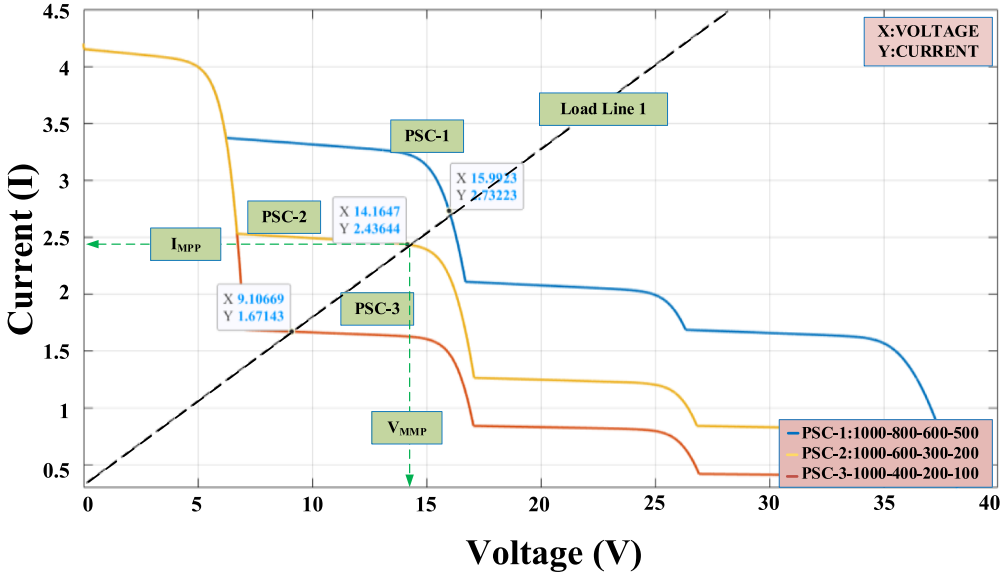


Fig. 4. Modeling the I-V Response of PV Modules to Dynamic Shading Conditions.

end to further exploration and stabilize the D. The global best power ($P_{Gbest}(k)$) is determined by selecting the highest power level from the individual best powers of each coot. Regarding the coot, the location is subsequently recognized as $D_{Gbest}(k)$. The search operation concludes when the discrepancy in the highest power value found globally between the previous $P_{Gbest}(k-1)$ and the current cycle's $P_{Gbest}(k)$ is below 5 %, and the positional difference between each coot is also below 5 %, as indicated by condition (15). If the condition is fulfilled, the variable D will stabilize with $D_{Gbest}(k)$ being utilized as the duty cycle value of the converter, this will also satisfy the termination condition. The requirement of ΔD being satisfied indicates that all four particles are near each other. Once the termination condition is reached, the values of (V_P), (I_{PV}), $D_{Gbest}(k)$, and $P_{Gbest}(k)$ will be referred to as V_{MPP}, I_{MPP}, D_{MPP}, and P_{MPP}, respectively.

$$\frac{|P_{Gbest}(k)-P_{Gbest}(k-1)|}{P_{Gbest}(k-1)} and \Delta D < 0.05 \tag{19}$$

Differentiation of irradiance and load variation.

To distinguish among these two scenarios and avoid re-initialization during load variation, the algorithm will first follow the GMPP using Eq. (20) and then observe the PV power ($P_{PV(k)}$) for any variation in power after the termination requirement for either the shaded or unshaded condition is reached. This indicates exposure to either irradiance change or load variation. The algorithm keeps using D at MPP if this requirement is not satisfied.

$$\frac{|P_{PV}(k)-P_{MPP}|}{P_{MPP}} > 0.05 \tag{20}$$

Fig. 4 illustrates the I-V curves during the various irradiances for three PV strings: PSC-A, PSC-B, and PSC-C. Initially, at point A, the MPP for PSC-A is tracked on the load line, yielding a (V_{MPP}) of 15.99 V and an (I_{MPP}) of 2.73 A. However, in the event of irradiance transitioning from PSC-A to PSC-B, the operational point on the I-V curve shifts from point A to B. This shift will automatically decrease the voltage and current values to 14.16 V and 2.43 A, respectively, instead of following the

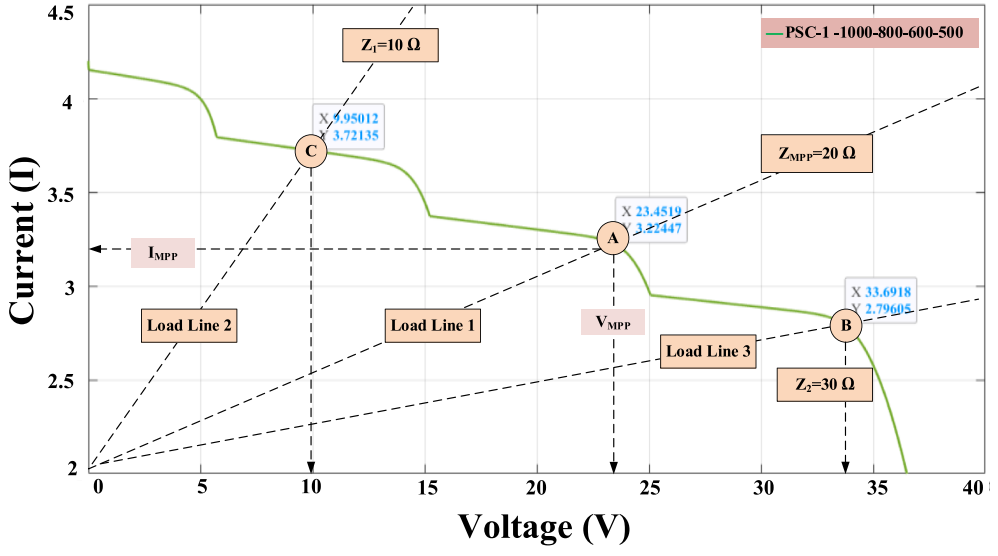


Fig. 5. The response of the I-V curve to changes in load conditions.

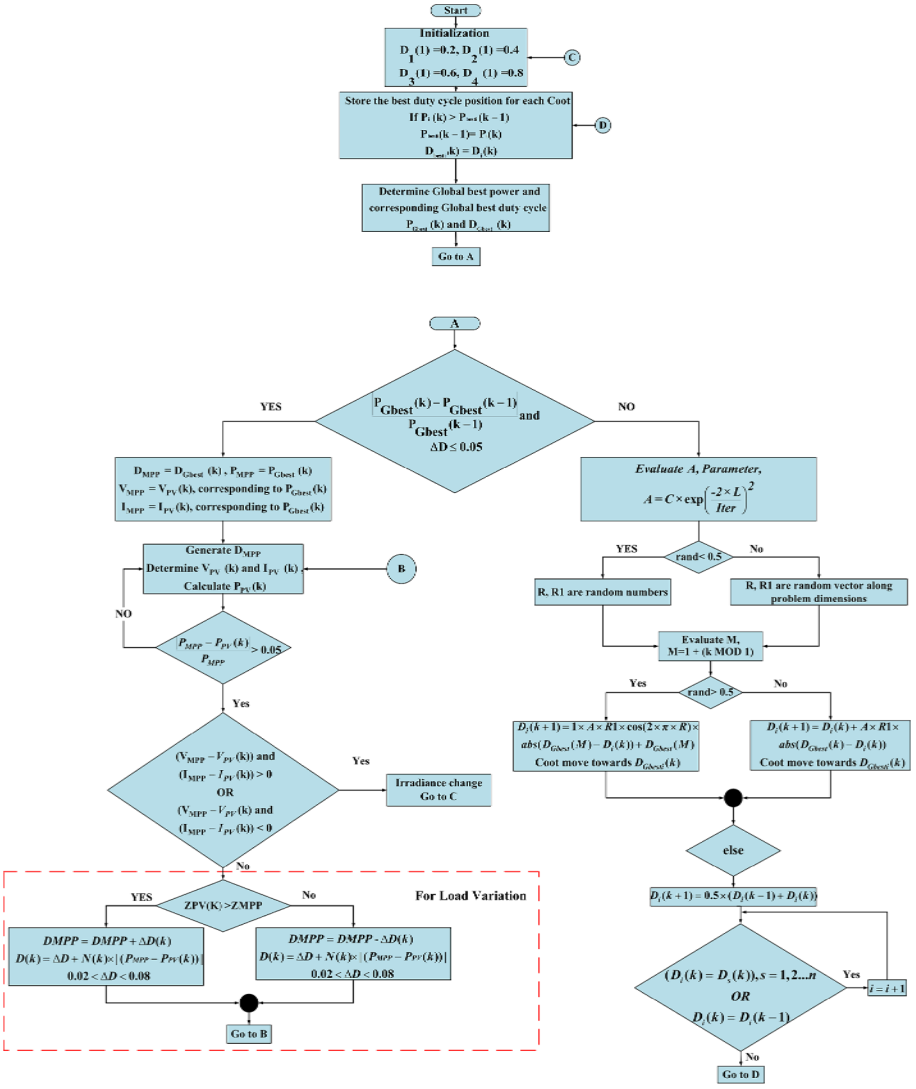


Fig. 6. The proposed algorithm flowchart.

tracked values under PSC A. On the other hand, the operational point on the I-V curve moves from point A to point C if the shadow condition changes from PSC-A to PSC-C. This shift leads to a decrease in both voltage and current (9.10 V and 1.67 A, respectively). Consequently, the stipulations outlined in Eq. (21) will be fulfilled when the system undergoes irradiance variations. Afterward, the algorithm will begin a re-initialization procedure to precisely track the new highest point in the system.

$$\begin{bmatrix} V_{PV} - V_{mpp} > 0 \\ AND \\ I_{PV} - I_{mpp} > 0 \end{bmatrix} OR \begin{bmatrix} V_{PV} - V_{mpp} < 0 \\ AND \\ I_{PV} - I_{mpp} < 0 \end{bmatrix} \tag{21}$$

Proposed approach for load variation.

In case the condition outlined in Eq. (21) is not fulfilled, it signifies the fulfillment of the stipulation articulated in Eq. (20) due to alterations in the load. This study presents a novel method to enhance the system's ability to adapt to variations in load and prevent the need to restart the algorithm from the beginning. The proposed approach employs a new method, as given in Eq. (22), to modify the tracking point when the load changes. The main goal is to eradicate consistent fluctuations and avert gradual deviations in the system.

$$\Delta D(k) = \Delta D + N(k) \times |(P_{MPP} - P_{PV(k)})| \tag{22}$$

where $N(k)$ is a scaling factor, $\Delta D(k)$ represents the step-size value given at each iteration, P_{MPP} indicates the power at the MPP, and $P_{PV(k)}$ is the power supplied by the PV panel. According to Eq. (22), the recommended adjustable step size perturbation (step) is automatically adjusted in this case. The modification is based on multiplying the photovoltaic power $P_{PV(k)}$ by a scaling factor (N) and the maximum power point MPP absolute value. To attain a fair trade-off between the decrease of steady-state fluctuations and the response time, the scaling factor (N) must be manually adjusted. It is crucial to remember that the scaling factor (N) must be selected to be less than 1 [35]. To illustrate the proposed method, as shown in Fig. 5, it is evident that when the system encounters variations in load after attaining the maximum power at point A and reducing the load value from load line 1 to load line 2, the operating point shifts from A to C. Over this period, the present value exhibits an increase while the voltage value decreases. Conversely, as the load increases from load line 1 to load line 3, the operating point transitions from A to B, leading to a rise in voltage and a decline in current. Based on the results of the experiments, the chosen range for $N(k)$ is 0.5 %. These results meet the specified step-size requirement, which is especially relevant when the operating point is very far from the maximum point. Moreover, the magnitude of the step must decrease

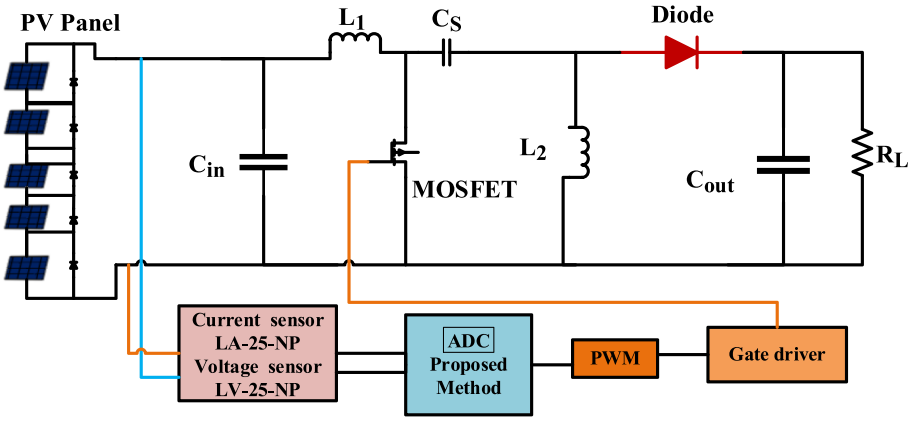


Fig. 7. Block diagram of the hardware setup (SEPIC converter).

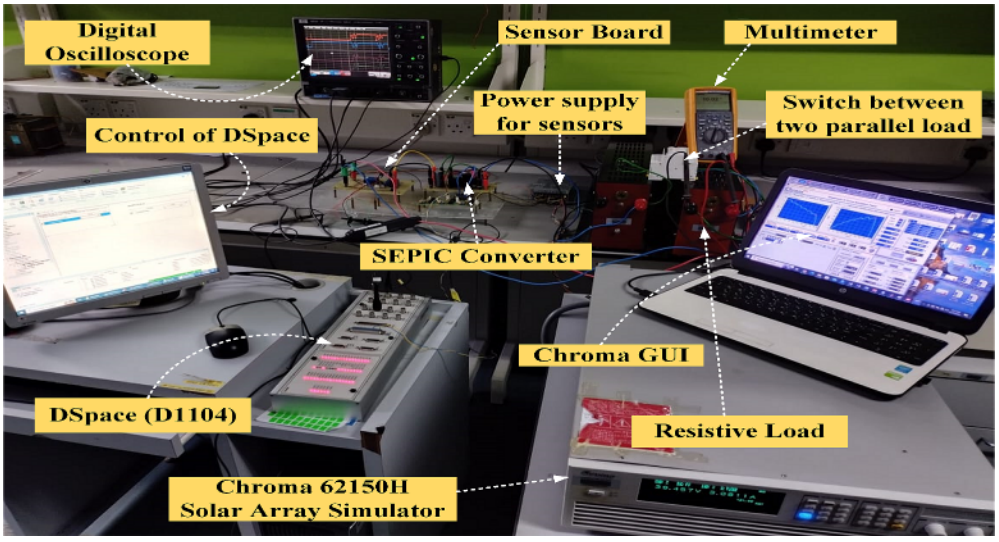


Fig. 8. Experimental setup.

gradually as the maximum power is approached. To address this, inconsequential step values that are too small to represent substantial disruptions are eliminated at each step. Instead, a perturbation with an effective minimum change of $\Delta D(k)\text{min} = 0.02$ is introduced. A maximum upper limit of $\Delta D(k)\text{max} = 0.08$ is enforced to prevent too large step values that might result in significant disruptions. Fig. 6 displays the flowchart of the proposed technique.

Results and discussion

The proposed ICOA is experimentally verified using a designed setup that includes a SEPIC converter, as shown in Fig. 7. The system parameters are specified as follows: input and output capacitances (C_{in} and C_{out}) set to 1000 μF, Cs (capacitor connected in series) at 330 μF, inductances L_1 and L_2 both at 470 μH, load resistance (R_{load}) maintained at 10 Ω, with a chosen MOSFET using a 25 kHz switching frequency. The selection of the DC-DC SEPIC converter holds paramount importance for two principal reasons. Firstly, it demonstrates the ability to seamlessly elevate or reduce its output voltage without encountering non-operational regions. This attribute ensures steadfast maintenance of output voltage polarity, a critical element in the proposed system. Secondly, and of heightened significance, it empowers the system to effectively track the GMPP at any position along the PV curve, whether positioned at the extremities or any intermediate point. To validate diverse shading scenarios involving a series connection of six modules and multiple peaks, as illustrated in Fig. 8, a Solar Array Simulator (SAS) is employed. Specifically, the Chroma Model 62150H is utilized to represent the PV array. The SAS is equipped with a graphical user interface (GUI), facilitating the precise localization of the tracking point within the P-V search area. A specialized GUI designed for compatibility with the SAS plays a dual role in testing and developing partial shading patterns. Subsequently, the partially shaded 200 W "Sun Power SPR-200-BLK U-2008 E" PV array transmits its generated power to the SEPIC through the SAS. The SAS software facilitates the selection of various partial shading patterns with specific parameters outlined in Table 2. The proposed methodology is executed under the control of DSpace (D1104) as the primary controller, ensuring a comprehensive and controlled experimental implementation. Voltage and current are detected for the DSpace ADC pin using the LV-25P voltage sensor and LA-25NP current sensor, both manufactured by LEM. The load being

Table 2
Sun Power SPR-200-BLK-U [2008 (E)] of 200 W under STCs.

(P_{MPP})	200 W
(V_{MPP})	40 V
(I_{MPP})	4.998 A
(V_{OC})	47.8 V
(I_{SC})	5.398 A

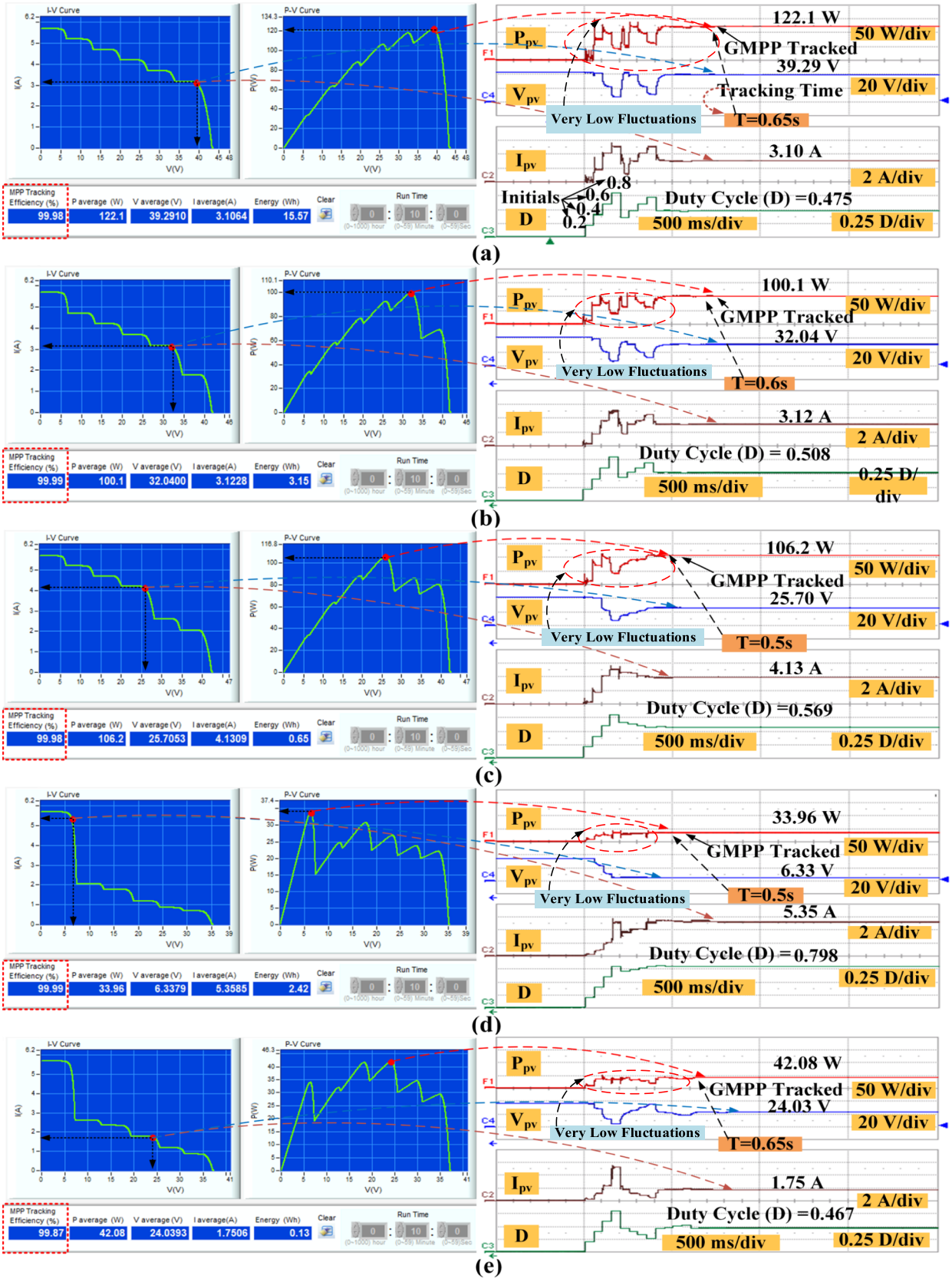


Fig. 9. Validation of the ICOA for (a) PSC-A to (e) PSC-E.

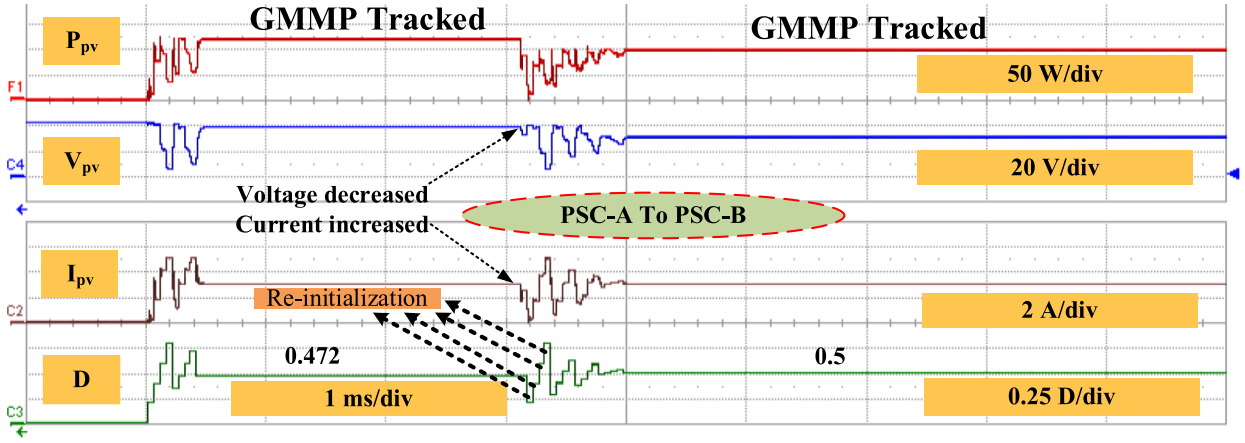


Fig. 10. Response to an irradiance change from PSC-A to PSC-B.

employed has a resistance of 10 Ω. The tracking and steady-state instances—P_{PV}, V_{PV}, I_{PV}, and D—are captured by the Lecroy Digital Signal Oscilloscope (DSO). The sample time of 0.05 s is used to guarantee that the DSpace controller has enough time to gather data from the voltage and current sensors.

Validation for different PSCs

Validation is conducted on the proposed technique on five PSCs, each exhibiting six distinct multiple peaks, as depicted in Fig. 9. The PV panel is configured with the following five complex PSCs:

1. PSC-A: 1000, 900, 800, 700, 600 and 500 W/m^2
2. PSC-B: 1000, 800, 700, 600, 500 and 250 W/m^2
3. PSC-C: 1000, 900, 800, 700, 400 and 300 W/m^2
4. PSC-D: 1000, 300, 250, 150, 100 and 75 W/m^2
5. PSC-E: 1000, 400, 350, 250, 150 and 100 W/m^2

In the (P-V) and (I-V) search spaces, the red dot represents the operational point of a photovoltaic array in a steady state. The GUI of the SAS, as shown in Fig. 8, presents the tracking efficiency calculated using Eq. (23), together with the P_{PV}, V_{PV}, and I_{PV} values.

$$MPPT\ efficiency = \frac{power\ of\ Steady\ state}{Maximum\ power\ available} \times 100 \quad (23)$$

Competing techniques are evaluated for their robustness through the examination of three distinct characteristics: tracking time (T), power output stability in the face of fluctuations, and efficiency.

In PSC-A, the GMPP is located far to the right in the PV exploration area, indicating an extractable power of 122.1 W, as shown in Fig. 9(a). The method showed the ability to track the global point with 99.98 % tracking efficiency and a quick convergence time of 0.65 s obtained with a steady state point of 0.475.

For PSC-B, varying irradiance levels of 1000, 800, 700, 600, 500, and 250 W/m2 are established, resulting in a maximum achievable power of 100.1 W. The GMPP is precisely identified with a rapid time of 0.6 s, attaining an outstanding tracking effectiveness of 99.99 %. The variable D exhibits stability, gradually converging towards a value of 0.508, as depicted in Fig. 9(b).

In PSC-C, the designated levels of irradiance are set at 1000, 900, 800, 700, 400, and 300 W/m2. These levels result in a significant maximum power output of 106.2 W. The GMPP was accurately tracked at 0.5 s, resulting in an outstanding tracking effectiveness of 99.98 %. In addition, D exhibited stability, converging to a constant value of 0.569, as represented in Fig. 9(c).

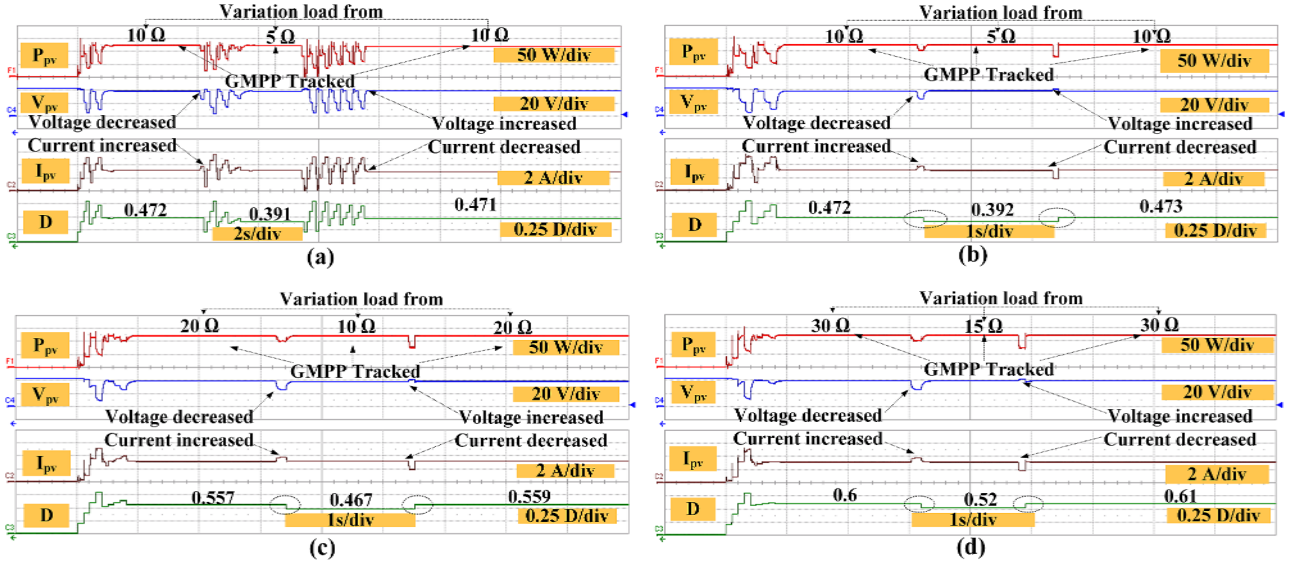


Fig. 11. Experimenting with load variation on (a) ICOA without proposed modification (10 to 5 Ω), (b) ICOA with the proposed modification (10 to 5 Ω), (c) (20 to 10 Ω), and (d) (30 to 15 Ω).

Table 3
Validation of ICOA for Load Variation.

APPLIED PSC	MPPT Time (s)	Conditions for Load variation	Recovery time for MPP (s)
PSC-A	0.45	(10–5) and vice versa	0.1
PSC-A	0.45	(20–10) and vice versa	0.1
PSC-A	0.45	(30–15) and vice versa	0.1

Regarding PSC-D, the designated irradiances are set at 1000, 300, 250, 150, 100, and 75 W/m2, leading to a maximum achievable power of 33.96 W. The *D* demonstrated stability, reaching a convergent value of 0.798. The GMPP was accurately monitored and tracked in just 0.5 s, providing a tracking efficiency of 99.99 %. This can be shown in Fig. 9 (d).

The irradiance in PSC-E was set at values of 1000, 400, 350, 250, 150, and 100 W/m2, as seen in Fig. 9(e), leading to a maximum achievable power of 42.08 W. In 0.65 s, the GMPP was tracked with 99.87 % efficiency. Concurrently, the D stabilized and was able to continuously converge to 0.467.

Furthermore, the ICOA technique performance under the specified conditions indicates that, irrespective of the shading pattern on the P-V characteristic, the PV system exhibits lower fluctuations during both steady-state and transient periods when implementing the ICOA–MPPT.

Verification for the dynamic change in MPP

To illustrate the efficacy of the proposed method during the dynamic irradiance change. Fig. 10 illustrates the moment when the irradiance transitioned from PSC-A to PSC-B. The GMPP was accurately tracked under the shading conditions of PSC-A. Subsequently, as the irradiance level transitioned from PSC-A to PSC-B, meeting the condition specified in Eq. (21), the algorithm underwent re-initialization and precisely tracked the GMPP for PSC-B.

Verification for load variation.

Experimental validation under varying loads involved two tests, as depicted in Fig. 11. Initially, with the conventional method, the algorithm re-initializes during load variance. Additionally, the proposed method employs an adaptive perturbation step size for precise MPPT tracking. The proposed method has been applied to loads ranging between (10–5), (20–10), (30–15), and vice versa under the same-circumstances in PSC-A to evaluate its effectiveness in managing load changes. The configuration of the switch breaker was placed between two parallel variable loads to change the load, as shown in Fig. 8. Using PSC-A and a resistive load of 10 Ω, the traditional technique tracked the GMPPT and then changed the load to 5 Ω. In this scenario, the algorithm re-initialized the tracking of the GMPPT and changed the load from 5 Ω to 10 Ω. To track the GMPP, the method will re-initialize, as shown in Fig. 11(a). In the proposed technique, the controller determines the power change due to load fluctuation during load variation between (10–5) and vice versa after monitoring the GMPP for PSC-A successfully, as shown in Fig. 11(b). Subsequently, the adaptive perturbation step-size technique is implemented, efficiently evaluating the GMPP within a time frame of 0.1 s. After successfully monitoring the GMPP, as shown in Fig. 11(c), the practice of revisiting the GMPP with incremental duty cycle modifications was accomplished in roughly 0.1 s for load changes ranging from 20 Ω to 10 Ω. During load variation between 30 Ω and 15 Ω, the GMPP tracking process acquired 0.1 s, as depicted in Fig. 11(d). Moreover, the proposed approach, illustrated in Fig. 11(b)-(d), reduced power oscillations during the acquisition of the GMPP value and improved the CS in response to load variations. Table 3 provides information regarding the duration of tracking for GMPP and the time needed to retrace the GMPP in the event of load changes.

Verification with simultaneous variation in load and dynamic irradiance change

The proposed method was implemented with simultaneous variation in load and dynamic irradiance change, as shown in Fig. 12. After tracking the MPP of PSC-A under a 10 Ω load, the system was subjected to a subsequent load variation, transitioning from 10 to 5 Ω at PSC-A. The results underscore the efficacy of the proposed technique in promptly and accurately tracking the operational point, thereby minimizing power fluctuations. Following this, the system underwent an irradiance transition from PSC-A to PSC-B at a resistance of 5 Ω. The approach demonstrated its proficiency in effectively and precisely monitoring the MPP under these conditions. Likewise, when the load was changed from 5 to 10 Ω at PSC-B, the approach showcased its adaptability and accuracy in tracking the MPP.

Comparative analysis

Comparative evaluation of performance between standard COA and ICOA

The conventional COA has been validated for PSC-A to PSC-E, highlighting the advantages of the proposed ICOA-MPPT. Through PSC-A to PSC-E, the P_{PV}, V_{PV}, I_{PV}, and *D* waveforms are depicted in Fig. 13, with the traditional COA technique for MPPT; Table 5 shows the tracking time and efficiency. The conventional algorithm required 2.5 s,

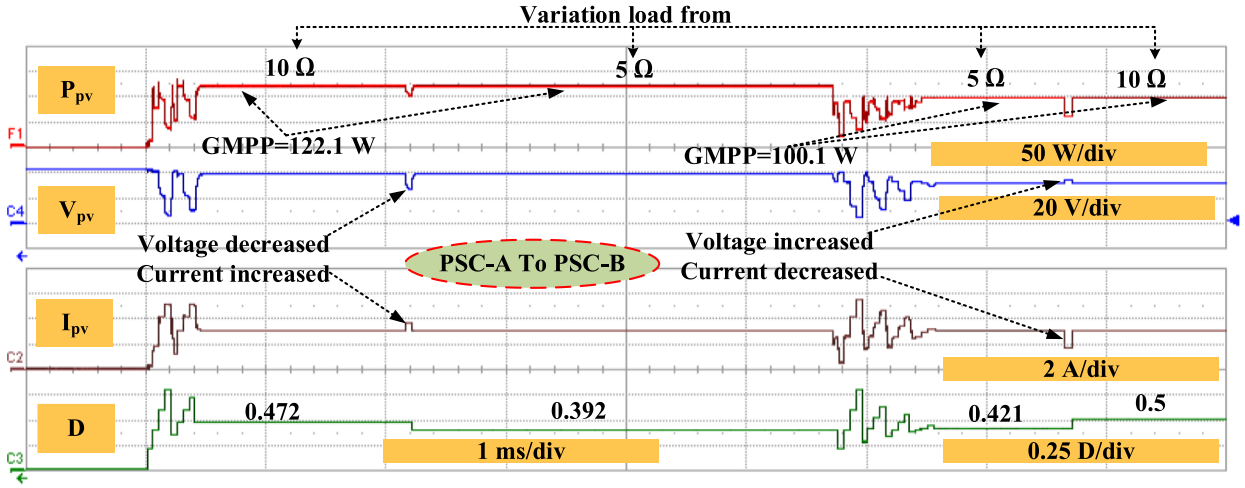


Fig. 12. Experimenting with load and irradiance variations.

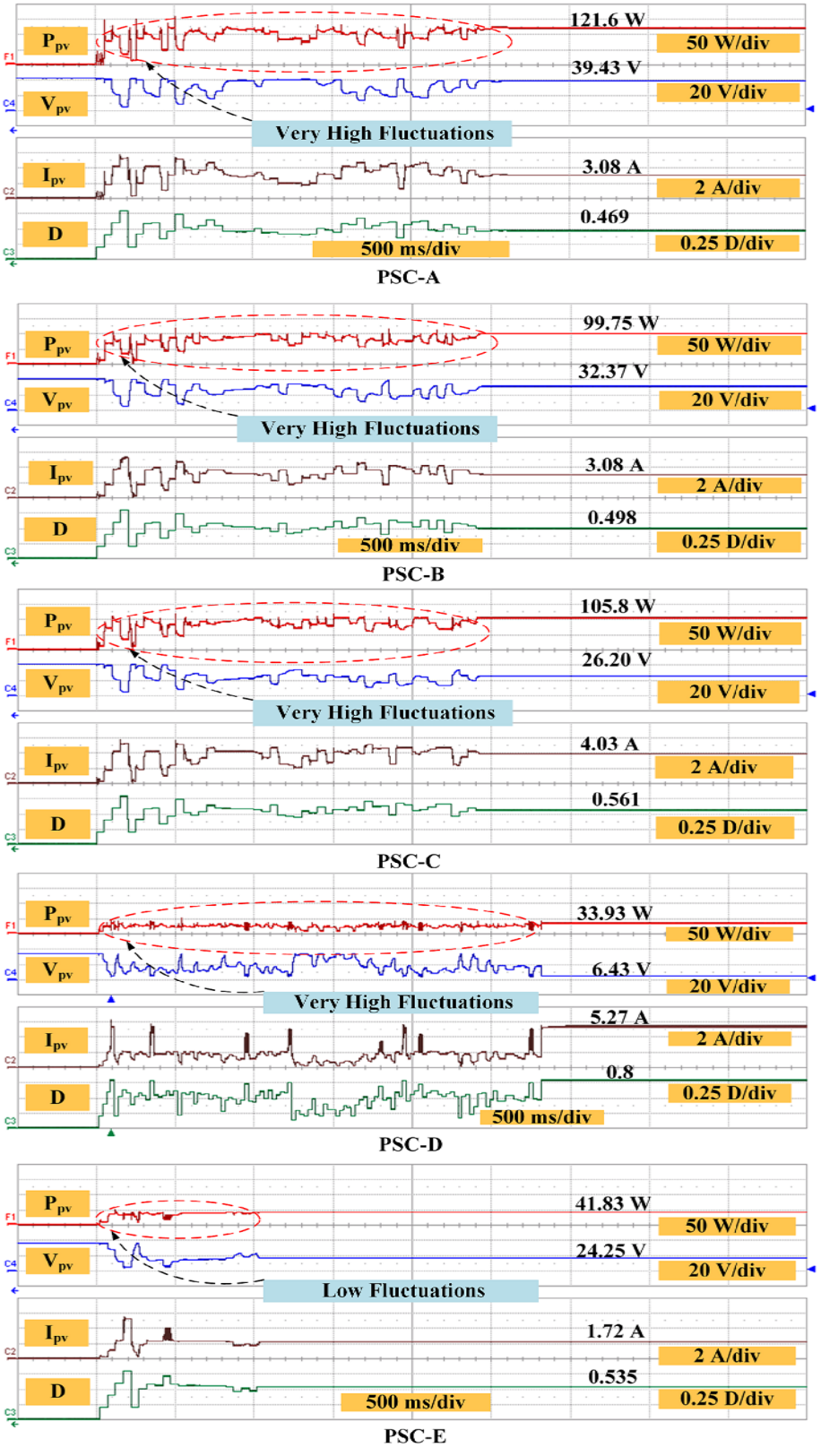


Fig. 13. Verification of experimental outcomes for the conventional COA across PSC-A to PSC-E.

2.4 s, 2.6 s, 2.7 s, and 1 s, respectively, to retrace its steps to the GMPP. By comparison, the suggested technique yielded faster tracking times of 0.65 s, 0.6 s, 0.5 s, 0.5 s, and 0.65 s, respectively, as shown in Fig. 9. The findings indicate that the ICOA has superior efficiency, less output fluctuation and quicker convergence than the GMPP in comparison to the traditional COA.

Comparative analysis using other metaheuristic approaches

The efficiency of tracking the GMPP relies significantly on the



Fig. 14. Experimental Comparison between (a) PSO, (b) IDE, (c) ITG, (d) ISSA, and (e) ISO from PSC-A to PSC-E.

effectiveness of various metaheuristic approaches. To assess the proposed approach's effectiveness, conducted a comparison with methods—PSO [13], IDE [30], ITG [28], ISSA [17], and ISO [32], under similar PSCs (A–E). Fig. 14(a)–(e) illustrates the outcomes of the experiments. The findings explicitly articulate the variations among the cases. Despite the relatively sluggish performance of CS compared to other algorithms, the PSO algorithm exhibited a precise monitoring capability for GMMP. In contrast, by reducing the use of random numbers, the IDE technique had a greater CS than the PSO algorithm. The ISO, ISSA, and ITG outperformed both the PSO and IDE, showcasing a unique dynamic tuning parameter and superior CS. Out of the several algorithms that were tested, ICOA was found to be the best performer. It achieved an

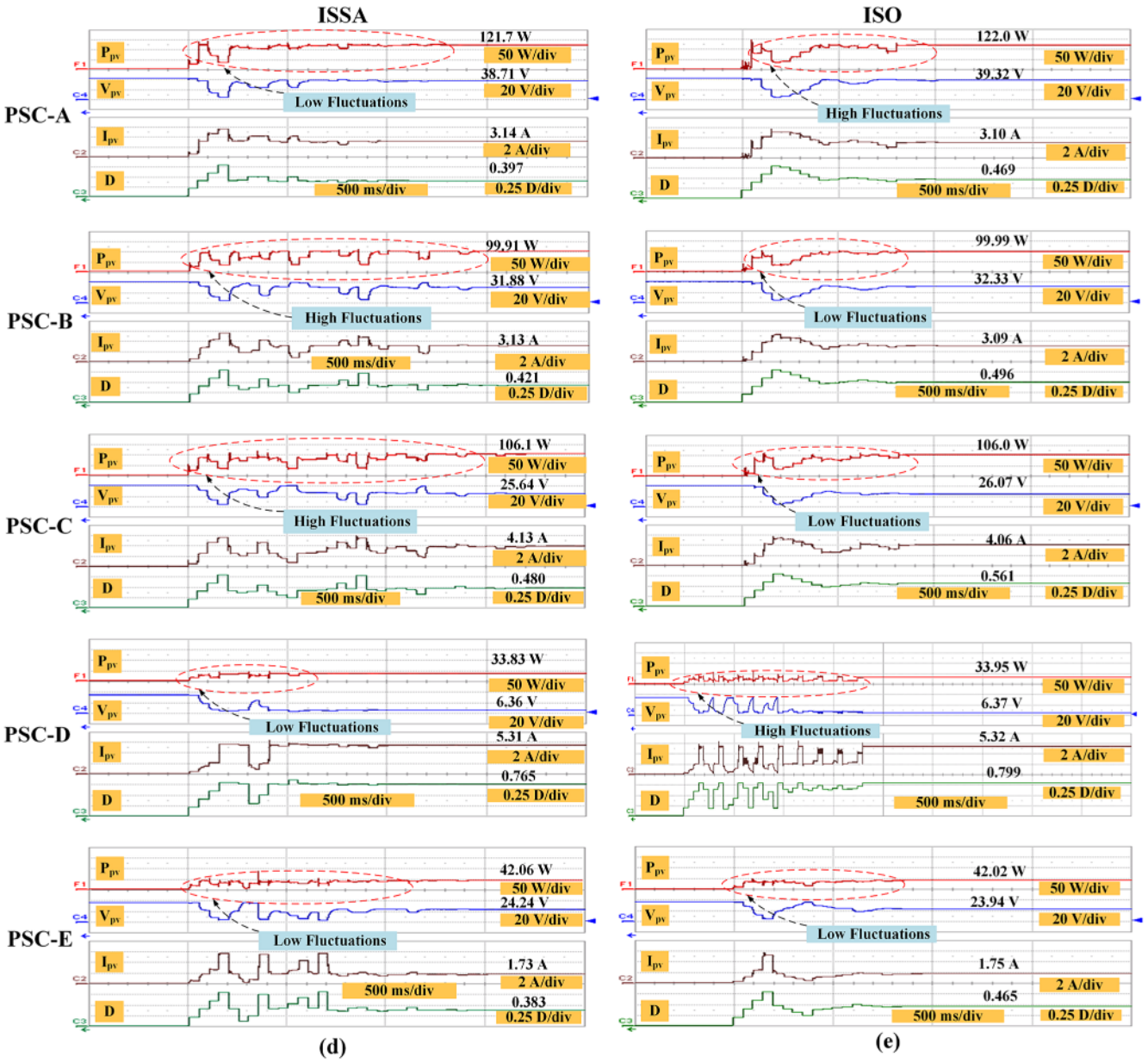


Fig. 14. (*continued*).

Table 4
Tuning parameters of PSO, IDE, ITG, ISSA, ISO, and ICOA.

Metaheuristic methods.	Tuning parameter
PSO [13]	$C_{1max=2}$, $C_{1min}=1$, $C_{2max}=2$, $C_{2min=1}$ $w_{max}=1$, $w_{min}=0.1$
IDE [30]	$F_{min}=0.1$, $F_{max}=0.4$, $CR=0.5$
ITG [28]	$r=(1-\frac{k}{kmax})$
ISSA [17]	$P_r=0.1$, $dg\in[0.5,1.11]$
ISO [32]	$c=(1-\frac{t}{tmax})$
ICOA	$A=C\times e^{-\left(\frac{2\times l}{I_{ter}}\right)^2}$

efficiency rate of 99.94 % and an average tracking time of 0.58 s for GMPP, as shown in Fig. 9 and Table 5. IDE attained an average tracking time of 2.08 s, operating at an efficiency rate of 99.71 %. ITG, on the other hand, recorded a tracking time of 1.22 s, with an efficiency rate of 99.74 %. ISSA exhibited a tracking time of 1.2 s, achieving an efficiency rate of 99.72 %. ISO demonstrated a tracking time of 1.02 s, with an efficiency rate of 99.87 %. On the other hand, PSO had the longest time for tracking, lasting 2.2 s, with an efficiency rate of 99.73 %. In general, the ICOA method proposed in the study exhibited enhanced performance in terms of tracking time, efficacy has lower fluctuation during GMPP tracking; thus, the stability of the system can be significantly increased when compared to existing metaheuristic algorithms. Table 4 provides a summary of the tuning parameters used for each algorithm.

A comparative analysis of load variation response

To demonstrate the effectiveness of the suggested approach in adjusting to variations in load circumstances, an experimental inquiry was carried out. The efficiency of the proposed technique was compared with two recently published studies, ITG [28] and IDE [30], to evaluate its effectiveness. The proposed method for load variations in IDE outperforms all previous metaheuristic techniques, quickly tracing the MPP for each load variation within 0.1 s, as Fig. 15(b) illustrates.

Table 5
The comparison of the ICOA approach with different metaheuristic methods.

Factors	MPPT with Bio-Inspired	PSO [13]	IDE [30]	ITG [28]	ISSA [17]	ISO [32]	Conventional COA	Proposed method
	No. of tuning parameters	3	3	1	2	1	3	1
	No. of randomly generated numbers	2	2	2	2	1	2	1
The average Tracking Time (s)	PSC-A	3.4	2.1	1.5	0.8	1	2.5	0.65
	PSC-B	2.7	1.6	1.3	1.5	0.85	2.4	0.6
	PSC-C	2.5	2.7	1.6	1.5	0.8	2.6	0.5
	PSC-D	0.7	1.9	0.7	0.9	1.65	5.7	0.5
	PSC-E	1.7	2.1	1	1.3	0.8	1	0.65
	Average	2.2	2.08	1.22	1.2	1.02	2.84	**0.58**
MPPT Efficiency (%)	PSC-A	99.48	99.93	99.84	99.65	99.91	99.63	99.98
	PSC-B	99.74	99.70	99.86	99.80	99.93	99.69	99.99
	PSC-C	99.77	99.85	99.91	99.86	99.79	99.60	99.98
	PSC-D	99.86	99.71	99.76	99.50	99.95	99.90	99.99
	PSC-E	99.76	99.32	99.33	99.77	99.73	99.28	99.87
	Average	99.73	99.71	99.74	99.72	99.87	99.62	**99.94**

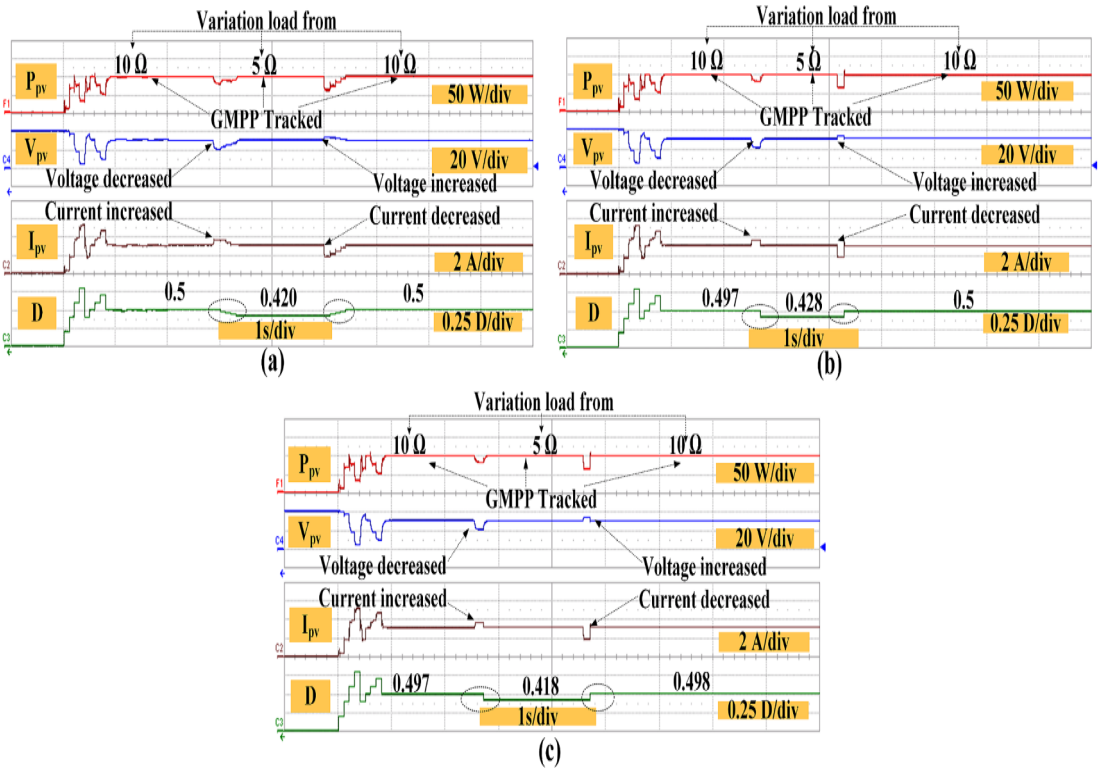


Fig. 15. Comparison of load variation between (a) ITG, (b) IDE, and (c) ICOA for PSC-B.

Nevertheless, a significant limitation of this method is its strong reliance on the specific converter employed, as the algorithm would necessitate substantial modifications if an alternative converter were utilized. Nevertheless, ITG used a specific method to trace the GMPPT by implementing a fixed perturbation *D* with a defined step size, as shown in Fig. 15(a). This specific technique required a time of 0.25 s. An inherent limitation of this strategy is that as the value of *D* increases, there is a higher likelihood of oscillation occurring around the GMPP. In contrast, decreasing the value of *D* extends the time needed to track the GMPP. In the face of changing load conditions, this paper introduces a novel approach that achieves rapid and precise tracking via adaptive perturbation step size. This strategy tries to mitigate steady-state oscillations and remove reliance on the selected converter without necessitating any adjustments to the algorithm, as illustrated in Fig. 15(c).

Conclusion

This paper proposes an improved coot optimization algorithm (ICOA) designed for implementing MPPT in solar power systems under different PSCs and load variations. The performance of the proposed MPPT method was evaluated and compared with the performance of other algorithms such as COA, PSO, IDE, ITG, ISSA, and ISO. The evaluation was conducted by experimental verification under five different complicated partial shading instances. The ICOA exhibited superior performance compared to the aforementioned approaches and consistently demonstrated excellent tracking capabilities in all scenarios, irrespective of the GMPP position. Furthermore, it exhibited enhanced efficiency in comparison to the COA, with a reduction of over 50 % in convergence time and a notable decrease in power fluctuations. The evaluated primary PSCs achieved efficiencies of an average of 99.94 %.

Furthermore, the proposed modification avoids re-initializing the system when it is exposed to a varying load, responding rapidly in 0.1 s with reduced 1. oscillations. The findings demonstrated the practicality of the suggested approach and its capacity to accurately monitor the GMPP with exceptional efficiency above 99.94 % in all tested scenarios, with the lowest average tracking time of 0.58 s. The future work will center on refining metaheuristic algorithms to effectively discern between partially and uniformly shaded patterns.

CRediT authorship contribution statement

Abdulbari Talib Naser: Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Karam Khairullah Mohammed:** Writing – review & editing. **Nur Fadilah Ab Aziz:** Supervision. **Karmila binti Kamil:** Supervision. **Saad Mekhilef:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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